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The effect of temperature on the adiabatic burning velocities of diluted hydrogen flames: A kinetic study using an updated mechanism

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ABSTRACT

The effect of temperature on the adiabatic burning velocities of diluted hydrogen flames has been analyzed using an updated version of the Konnov detailed reaction mechanism for hydrogen. The contemporary choice of the reaction rate constants is provided with the emphasis on their uncertainties, and the analysis of the performance of the updated mechanism is presented and compared to the previous version for a wide range of validation cases: jet stirred and flow reactors; oxidation, decomposition and ignition in shock waves; ignition in rapid compression machines; laminar burning velocity and flame structure. An overall improvement of the mechanism performance was observed, particularly for the shock tube and flow reactor studies. Temperature dependence of the burning velocity, S_L , is commonly interpreted using the correlation $S_L = S_{L0} (T/T_0)^\alpha$. The updated mechanism was applied to study the behavior of the power exponent α for $H_2 + O_2 + N_2$ flames in a wide range of stoichiometry and dilution ratios. The simulations were compared to the available experimental results, either taken from the literature or evaluated in the present study from the existing burning velocity data. The equivalence ratio and N_2 content in the mixture were found to have significant influence on the temperature power exponent. The dependence of the temperature exponent on the fitting temperature range was observed and discussed. This effect was found to cause significant discrepancies in the burning velocities at high temperatures, if obtained with empirical correlation.

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1. Introduction

Hydrogen-fueled internal combustion engines and stationary gas turbines in integrated gasification combined cycle (IGCC) technology burning high hydrogen content syngas mixtures are nowadays the most realistic and promising examples of hydrogen-based economy. Further development and optimization of these technologies requires detailed knowledge of combustion characteristics of hydrogen at elevated pressures and temperatures. The laminar burning velocity is one of these characteristics that defines the rate of fuel consumption in laminar flames, and also relates to blow-off and flash-back phenomena. Moreover, contemporary models of turbulent flame propagation are calibrated by the laminar burning velocity at specific (local) conditions of composition, temperature, pressure, and dilution by products. Since the laminar burning velocity depends on the limited number of these parameters it

can be described by mathematically simple correlations valid over the desired range of conditions.

Very often the effects of composition, temperature, pressure, and dilution by products are considered to be independent. For instance, the correlation describing the influence of initial temperature on the adiabatic laminar burning velocity and used in this or equivalent form since 1950s, e.g., [1],

$$S_L = S_{L0} (T/T_0)^\alpha \quad (1)$$

includes the reference temperature, T_0 , and the burning velocity at this temperature, S_{L0} , varying with equivalence ratio, ϕ . Pressure dependence of the power exponent coefficient, α , in Eq. (1) was often neglected or averaged. Earlier measurements of the power exponent α for hydrogen + air flames [1–4] at pressures around 1 atm were summarized by Konnov [5]. These experimental results have been compared with the power exponents derived from the modeling employing detailed reaction mechanism for hydrogen combustion [6]. Satisfactory agreement was found with the results of Heimeil [1] and with the correlations of Liu and MacFarlane [2] and of Iijima and Takeno [4] from stoichiometric to moderately rich ($\phi = 3$) mixtures. In the lean and very rich mixtures, however, the

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calculated power exponent increases dramatically and significantly deviates from the linear type correlations [2,4]. Discrete measurements of Verhelst et al. [7] in lean hydrogen flames at atmospheric pressure showed large scattering as can be seen in Fig. 1. Hu et al. [8] performed limited set of experiments at elevated temperatures and supported their measurements by the modeling employing the mechanism of Ó Conaire et al. [9]. For stoichiometric hydrogen + air flames at 1 atm they found that the calculated power exponent α increases with the temperature increase from 300 to 900 K and described this variation as $\alpha = 1.319 + 8.019 \cdot 10^{-4} \cdot T$, which gives the value of 1.56 at 300 K also shown in Fig. 1. Recent experimental study of Gerke et al. [10] covered a range of temperatures from 350 to 700 K and pressures from 5 to 45 atm. The authors derived empirical correlation for the pressures above 10 atm and equivalence ratios $0.4 < \phi < 2.5$ as $\alpha = 0.0163/\phi + 2.2937$. This expression is not shown in Fig. 1 since, as will be discussed below, the power exponents do depend on the pressures. Similar to the work of Hu et al. [8], Gerke et al. [10] implemented the mechanism of Ó Conaire et al. [9] and derived a correlation covering extended range of pressures ($1 < P < 80$ atm); the calculated power exponents from both studies are also depicted in Fig. 1. Contrary to the finding of Hu et al. [8] their correlation for α does not depend on temperature.

Rakopoulos et al. [11] compared correlations derived from experiments available in the literature and found that they are significantly different at conditions relevant to hydrogen-fueled engines, i.e., they predict laminar burning velocities diverged by a factor of two. Further experiments on hydrogen flame propagation are therefore highly desired. Meanwhile several groups derived burning velocity correlations thanks to the availability of detailed kinetic models for hydrogen combustion. D'Errico et al. [12] employed the reaction scheme developed by Frassoldati et al. [13] and calculated burning velocities of lean hydrogen + air flames from atmospheric pressure up to 16 atm. Bougrine et al.

[14] attempted several models and selected GRI-mech 3.0 [15] for the modeling of methane + hydrogen flames in the full range of fuel compositions up to 100% of H_2 . Verhelst and Sierens [16] chose the mechanism of Yetter et al. [17] for S_L correlation at 1–16 atm., and in the later study of Verhelst et al. [18] the model developed by Konnov [6] was used in the pressure range of 5–45 atm. Szabo et al. [19] selected rather old mechanism of Maas and Warnatz [20] integrated into the code INSFLA. The study of Ravi and Petersen [21] was motivated by the recent appearance of an updated detailed kinetic model for hydrogen and syngas combustion [22]. They clearly demonstrated that the power exponents α increase with the pressure increase and discussed different approaches of simultaneous description of the burning velocity variation with temperature and pressure. Thus, for the fair comparison, in Fig. 1 the power exponents α from these works [8,10,12,14,16,18,21] are shown if they were computed at 1 atm. Kusharin et al. [23] studied expanding spherical flames experimentally and numerically, and their value of α from the modeling with the mechanism of Warnatz [24] for H_2 + air mixture is also shown in Fig. 1.

The influence of diluents added into combustible mixture on the burning velocity is considered to be independent from the temperature dependence with typical linear correlation term:

$$S_L = S_{L0}(T/T_0)^{\alpha}(1 - F(\phi, T_0, \text{diluent}) * f) \quad (2)$$

where dilution ratio f is defined as the volume of the diluent divided by the total volume of the mixture (diluent + air + fuel). The coefficient of proportionality F representing the slope of corresponding dependences is known to be different for different diluents, e.g., [25,26]. It is now well established that it also depends on the initial temperature and stoichiometric ratio, e.g., [27,28].

The variation of the power exponents α with increased amount of steam or nitrogen added to stoichiometric hydrogen flames has been first studied by Kusharin et al. [23] at pressures around 1.5 atm. They observed that increased amount of diluent makes the flames more sensitive to the initial temperature of the mixture and thus the power exponent increases. This finding is consistent with the work of Zitoun and Deshaies [29] who investigated hydrogen + oxygen (no diluent) flames and for stoichiometric mixture found $\alpha = 0.9$, the value which is much lower than those shown in Fig. 1 for H_2 + air flames. Zitoun and Deshaies [29] also admitted that their burning velocity measurements are systematically 20% higher than earlier results of Edse and Lawrence [30], yet noted good agreement in terms of the temperature dependence. Increase of the power exponent coefficient α with dilution by nitrogen has been also observed by Vancoillie et al. [28] in methanol + oxygen + nitrogen mixtures.

Most recently Paidi et al. [31] performed a dedicated study of the effect of dilution by N_2 or CO_2 on the laminar burning velocity of hydrogen flames at elevated temperatures up to 600 K. Two dilution ratios defined as the diluent volume over the volume of (diluent + fuel) in the mixtures were studied. It was demonstrated that the power exponent coefficients α increase at higher dilution ratio. These experiments were found in satisfactory agreement with the modeling performed using the mechanism of Li et al. [32]. Experimentally obtained coefficients α , however, progressively deviate from the calculated values in rich mixtures [31].

There are two apparent reasons for significant differences in the power exponent coefficients α predicted by different detailed kinetic schemes as seen in Fig. 1. First, the modeling has been performed over different ranges of pressures and temperatures that affect pressure-dependent coefficients α as mentioned above. Moreover, in the correlation studies, the validation domain is centered at the engine-like conditions, so that if a single expression for α was obtained for the whole pressure range, it can be expected to

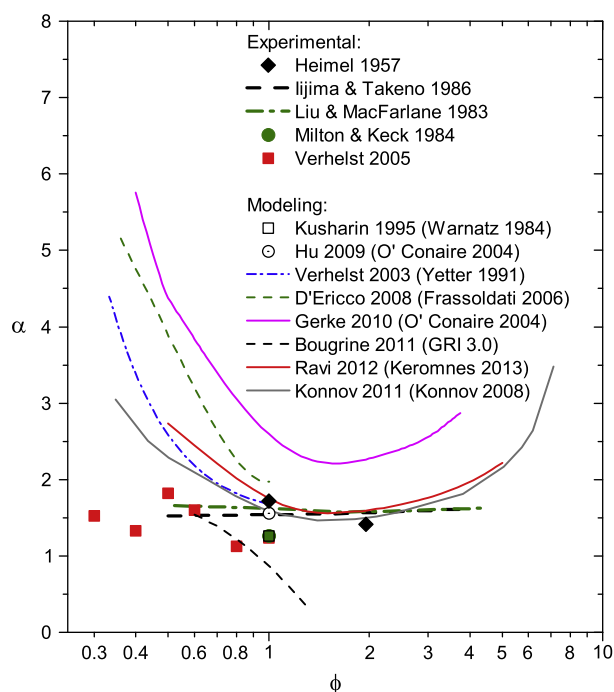


Fig. 1. Power exponent α for hydrogen + air flames at conditions close to atmospheric pressure and 298 K. Solid symbols and thick lines: experiments [1–4,7], open symbols and thin lines: modeling and correlations [5,8,10,12,14,16,21,23]. For the modeling studies, the implemented kinetic mechanisms [6,9,13,15,17,22,24] are given in the parentheses in the legend.

have a bigger discrepancy at the border condition of 1 atm. This can be seen from Fig. 1 comparing the modeling of Hu et al. [8], performed at 1 atm, and the results of Gerke et al. [10], obtained with the same kinetic scheme, but correlated for the pressure range of 1–80 atm. Second, even more important, different models have different predictive abilities even if they were validated against similar or the same commonly accepted sets of laboratory experiments, as it was emphasized by Burke et al. [33]. Independent comparison of detailed kinetic schemes for hydrogen combustion clearly reveals these differences as demonstrated by, e.g., Sepman et al. [34], Weydahl et al. [35] and Bezgin et al. [36]. Weydahl et al. [35] attempted direct comparison of the most important reactions in the following models: GRI-mech [15], Davis et al. [37], Konnov [6], Li et al. [32], Ó Conaire et al. [9], Williams [38], and Leeds [39]. Their compilation indicates that many recent mechanisms often share the same rate constants from 2 to 3 original sources, yet in different combinations that makes them distinct.

More systematic and extensive analysis of the performance of 19 recent hydrogen combustion mechanisms has been performed by Olm et al. [40]. These mechanisms have been tested in comparison with ignition delays in shock waves and rapid compression machines, burning velocity measurements and concentration–time profiles in flow and jet-stirred reactors, covering wide ranges of temperature, pressure and equivalence ratio. The analysis indicates that the performance of the mechanism of Kéromnès et al. [22] is currently the best, but several other mechanisms have similarly good overall performance. This short list includes the models of Saxena and Williams [41], Konnov [6], Hong et al. [42], 2 models originating from Princeton [32,33] and from NUIG [9,22].

In the present work, the rate constants implemented in the Konnov mechanism [6] are compared with those from the other models published since 2008 [22,33,42,43]. It is demonstrated in the following that consensus on at least some of the hydrogen reactions can be reached often thanks to recent careful measurements of Hong et al. [44–48]. Based on this analysis, a new mechanism for hydrogen combustion is proposed and then validated in comparison with commonly accepted sets of laboratory experiments.

Furthermore the temperature dependence of the laminar burning velocity of hydrogen flames is investigated in the wide range of dilution by N_2 from pure $H_2 + O_2$ up to highly diluted mixtures. The variation of the power exponent coefficients α with the dilution ratio is then analyzed and discussed.

2. Reaction mechanism

2.1. Species, thermodynamic and transport parameters

The models considered in the present work [6,22,33,42,43] share the same list of species. Rare recent mechanisms, e.g., [36], include reactions of ozone even though Dougherty and Rabitz [49] demonstrated that pertinent reactions are not important over all conditions of hydrogen oxidation covering three explosion limits. Likewise vibrationally or electronically excited species are not considered in the contemporary combustion models, although some disputable arguments were put forward in the literature, e.g., [50].

Thermodynamic data used in the present model are all from the latest database of Goos et al. [51].

The Chemkin package, software commonly used for combustion modeling, was traditionally accompanied by the transport properties database originating from Sandia National Laboratories [52] and interpreted by TRANFIT code. Paul and Warnatz [53] realized that simple description of transport properties could and should be improved and proposed a new transport model, DRFM. The

problem of accurate calculation of the binary diffusion coefficients was re-visited by Middha et al. [54] who demonstrated that TRANFIT underpredicts diffusion coefficients as compared to first-principle calculations. Wang et al. [55] then developed a modified interpreter that can be combined with the Chemkin package. This interpreter has been implemented by Burke et al. [33] who noted that similar agreement with their validation set (laminar burning velocities of premixed flames) is achieved using conventional TRANFIT code. Dong et al. [56] emphasized that transport properties are very important in predicting extinction phenomena in premixed flames, while Esposito and Chelliah [57] concluded that uncertainties in binary diffusion in simulation of premixed and non-premixed flames of hydrogen is of inferior importance as compared to chemical kinetics involved. Brown et al. [58] reviewed different approaches for predicting transport properties for combustion modeling and found that combination of the TRANFIT code with recently measured diffusion coefficients for OH, HO_2 , and ozone [59] leads to improved description of diffusion at least for OH. Therefore the Lennard-Jones parameters proposed by Ivanov et al. [59] are adopted in the present model.

2.2. Reactions

The detailed reaction mechanism used in this study is listed in Table 1. The sources of the rate constants implemented in other models [6,22,33,42,43] are also shown. All reactions are reversible; in the modeling, the reverse rate constants were calculated from the forward rate constants and thermodynamic data by the Chemkin interpreter code [52]. When reaction is written in the reverse direction the source of its rate constant is followed by the notation “rev”. The rate coefficients in the present work are given in $cm^3 - mole^{-1} - s$ units, while activation energies are in cal/mole. In the following, the sources of the rate constants are outlined and particular choices are discussed. Also temperature ranges over which the rate constants were determined and associated uncertainty are presented. An estimated uncertainty factor, UF, implies that the rate constant is expected to be in the range $k/UF < k < k * UF$. This simplified description of the uncertainties is mostly an indication of the possible maximum deviation of the “real” rates from the expressions adopted in the mechanism. Improved presentation of the uncertainty information has been discussed by Nagy and Turanyi [60] and others. It was demonstrated [61] that dedicated optimization procedures may bring significantly reduced uncertainties varying with temperature and lead to an improvement in predicting ability [62].

The rate constants of reactions (R1), (R2), (R3), (R7), (R9), (R19) and (X3) have not been changed from the previous version of the mechanism [6]; their choice is outlined in the Supplemental material. Although the rate constants of reactions (R4) and (R16) were also kept the same, they are described in the following to illustrate modifications to the original expressions from literature implemented in some mechanisms, e.g. by Keromnes et al. [22].

A large uncertainty of the rate constant of water decomposition

$$H_2O + M = H + OH + M \quad (R4)$$

was discussed by Li et al. [32] who proposed modification of the pre-exponential factor to improve burning velocity predictions at higher pressures. Srinivasan and Michael [63] investigated water decomposition within 2196–2792 K range for Ar and H_2O bath gases and derived rate constants adopted in the present work and by Hong et al. [42] and Burke et al. [33]. Subsequent high-level quantum chemistry calculations of this rate constant [64] significantly differ from these measurements [63] predicting higher rates at high temperatures and lower rates at room temperature as compared to recommendations of Baulch et al. [65]. Shimizu et al. [43] proposed slight modification (by 8%) to the rate constant of Li et al.

Table 1
H/O kinetic mechanism, units are $\text{cm}^3 - \text{mole} - \text{s} - \text{cal} - \text{K}$, $k = A T^n \exp(-E_a/RT)$. Notation: UF – uncertainty factor, “=” – the same rate constant as in the current mechanism, “0” – absent, “rev” – reverse reaction.

No.	Reaction	A	n	E_a	Temperatures	UF	Source	Konnov [6]	Shimizu et al. [43]	Hong et al. [42]	Burke et al. [33]	Keromnes et al. [22]
1a	$\text{H} + \text{H} + \text{M} = \text{H}_2 + \text{M}^a$	7.00E+17	−1.0	0	77–5000	2	[114]	=	=	[110] rev	[110] rev	[110] rev
	Enhanced third-body efficiencies (relative to Ar):				300–2000	5	[114]	=	=	=		
	$\text{H}_2 = 0, \text{N}_2 = 0, \text{H} = 0, \text{H}_2\text{O} = 14.3$								$\text{O}_2 = 2.2$			
1b	$\text{H} + \text{H} + \text{H}_2 = \text{H}_2 + \text{H}_2$	1.00E+17	−0.6	0	50–5000	2.5	[114]	=	=	[67] rev		
1c	$\text{H} + \text{H} + \text{N}_2 = \text{H}_2 + \text{N}_2$	5.40E+18	−1.3	0	77–2000	3.2	[114]	=	=	[110] rev		
1d	$\text{H} + \text{H} + \text{H} = \text{H}_2 + \text{H}$	3.20E+15	0	0	50–5000	3.2	[114]	=	=			
2	$\text{O} + \text{O} + \text{M} = \text{O}_2 + \text{M}^a$	1.00E+17	−1.0	0	300–5000	2	[24]	=	[110]	[110]	[110]	[110]
	Enhanced third-body efficiencies (relative to Ar):								[110]	[110]	[110]	[110]
	$\text{O} = 28.8, \text{O}_2 = 8, \text{N}_2 = 2, \text{H}_2\text{O} = 5$					2	[115,116]	=				
						3	[117]	=				
3	$\text{O} + \text{H} + \text{M} = \text{OH} + \text{M}^a$	6.75E+18	−1.0	0	2950–3700	3	[118]	=	=	[110]	[110]	[110]
	Enhanced third-body efficiency: $\text{H}_2\text{O} = 5$						[119]	=	=			
4a	$\text{H}_2\text{O} + \text{M} = \text{H} + \text{OH} + \text{M}^a$	6.06E+27	−3.312	120,770	300–3400	2	[63]	=	[32] * 0.92 rev	=	=	[43] rev
	Enhanced third-body efficiencies (relative to Ar):						[63]	=		=	=	
	$\text{H}_2\text{O} = 0, \text{H}_2 = 3, \text{N}_2 = 2, \text{O}_2 = 1.5, \text{He} = 1.1$											
4b	$\text{H}_2\text{O} + \text{H}_2\text{O} = \text{H} + \text{OH} + \text{H}_2\text{O}$	1.00E+26	−2.44	120,160	300–3400	2	[63]	=	[32] * 0.92 rev	=	=	[43] rev
5a	$\text{H} + \text{O}_2(+\text{M}) = \text{HO}_2(+\text{M})^{a,b}$	4.66E+12	0.44	0	300–2000	1.2	[66]	=	See text	See text	=	=
	Low pressure limit:	1.74E+19	−1.232	0	300–700	1.3	[72]	[66]	=	[70]	=	=
	$F_{\text{cent}} = 0.67$											
	Enhanced third-body efficiencies (relative to N_2):						[72]					
	$\text{Ar} = 0, \text{H}_2\text{O} = 0, \text{O}_2 = 0, \text{H}_2 = 1.5, \text{He} = 0.57$											
5b	$\text{H} + \text{O}_2(+\text{Ar}) = \text{HO}_2(+\text{Ar})^b$	4.66E+12	0.44	0	300–2000	1.2	[66]	=	See text	See text	=	=
	Low pressure limit:	4.57E+18	−1.12	0	300–700	1.3	[72]	[66]	=	[70]	=	[70]
	$F_{\text{cent}} = 0.5$											
5c	$\text{H} + \text{O}_2(+\text{O}_2) = \text{HO}_2(+\text{O}_2)^b$	4.66E+12	0.44	0	300–2000	1.2	[66]	=	See text	See text	=	=
	Low pressure limit:	5.69E+18	−1.094	0	300–700	1.3	[72]	=	[43]	=	=	
	$F_{\text{cent}} = 0.5$											
5d	$\text{H} + \text{O}_2(+\text{H}_2\text{O}) = \text{HO}_2(+\text{H}_2\text{O})^b$	4.66E+12	0.44	0	300–2000	1.2	[66]	[70]	See text	See text	=	=
	Low pressure limit:	3.67E+19	−1.0	0	1050–1250		[70]	=	[67]	=	=	
	$F_{\text{cent}} = 0.8$											
6	$\text{H}_2\text{O}_2(+\text{M}) = \text{OH} + \text{OH}(+\text{M})$	2.00E+12	0.9	48,750	500–1500	1.5	[78]	[6] rev	[67]	[77]	=	=
	Low pressure limit:	2.49E+24	−2.3	48,750	500–1500	2.5	[78]	[67] rev	[76]	[45]	=	=
	$F_{\text{cent}} = 0.42$											
	Enhanced third-body efficiencies (relative to Ar):						[78]				=	=
	$\text{H}_2\text{O} = 7.5, \text{H}_2\text{O}_2 = 7.7, \text{O}_2 = 1.2, \text{N}_2 = 1.5, \text{He} = 0.65, \text{H}_2 = 3.7$											
7	$\text{O} + \text{H}_2 = \text{OH} + \text{H}$	5.08E+04	2.67	6292	297–2495	1.3	[120]	=	=	[67]	[67]	=
8	$\text{H} + \text{O}_2 = \text{OH} + \text{O}$	1.04E+14	0	15,286	1100–3370	1.1	[46]	[67]	[80]	=	=	=
9	$\text{H}_2 + \text{OH} = \text{H}_2\text{O} + \text{H}$	2.14E+08	1.52	3450	300–2500	1.5	[67]	=	[121]	=	[121]	[122]
10	$\text{OH} + \text{OH} = \text{H}_2\text{O} + \text{O}$	3.34E+04	2.42	−1930	250–2400	1.5	[67]	=	[86]	[83]	=	[84] rev
11	$\text{HO}_2 + \text{O} = \text{OH} + \text{O}_2$	2.85E+10	1	−723.9	150–1600	2	[33]	[89]	[65]	[89]	[90] * 0.6	[65]
12	$\text{H} + \text{HO}_2 = \text{OH} + \text{OH}$	7.08E+13	0	300	300–1000	2	[81]	[6]	=	=	=	=
13	$\text{H}_2\text{O} + \text{O} = \text{HO}_2 + \text{H}$	2.20E+08	2.0	61,600	1500–4000	5	[99]	[67] rev	[43] rev	[67] rev	0	0
14	$\text{H}_2 + \text{O}_2 = \text{H} + \text{HO}_2$	7.40E+05	2.43	53,500	400–2300	1.5	[102]	[67] rev	=	=	[102] * 0.75 rev	[102] * 0.7
15	$\text{HO}_2 + \text{OH} = \text{H}_2\text{O} + \text{O}_2^c$	7.0E+12	0	−1093	300–2200	2	[48]	[67]	[65]	[47]	[103]	[103] * 0.85
	+	4.5E+14	0	10,930			[48]	[67]				
16	$\text{HO}_2 + \text{HO}_2 = \text{H}_2\text{O}_2 + \text{O}_2^c$	1.03E+14	0	11,040	300–1250	2.5	[76]	=	[107]	[107]	[107]	[107] * 0.87
	+	1.94E+11	0	−1409		1.4	[76]	=	[92]	[107]	[107]	[107]
17	$\text{H}_2\text{O}_2 + \text{H} = \text{HO}_2 + \text{H}_2$	5.02E+06	2.07	4300	300–2400	3	[113]	[67]	=	[15]	[110]	=
18	$\text{H}_2\text{O}_2 + \text{H} = \text{H}_2\text{O} + \text{OH}$	2.03E+07	2.02	2620	300–2400	3	[113]	[67]	=	[67]	[110]	[110]
19	$\text{H}_2\text{O}_2 + \text{O} = \text{HO}_2 + \text{OH}$	9.55E+6	2	3970	300–2500	3	[110]	=	[65]	[67]	=	=
20	$\text{H}_2\text{O}_2 + \text{OH} = \text{HO}_2 + \text{H}_2\text{O}^c$	1.74E+12	0	318	280–1640	1.5	[45]	[104]	[104]	=	=	=
	+	7.59E+13	0	7269		1.5	[45]	[104]	[104]	=	=	=

^a All other species have efficiencies equal to unity.

^b The fall-off behavior of this reaction is expressed in the form as used by Baulch et al. [67] and others.

^c Rate constant is the sum of two expressions.

[32], which was also adopted by Keromnes et al. [22]. In the present model collisional efficiency for He = 1.1 is introduced following Burke et al. [33].

Temperature and pressure dependence of recombination



is a focus of several recent experimental and theoretical studies aiming at improved accuracy in interpretation of available experimental data. However, different high-pressure rate constants are still in use in different models. For example, the rate constant of Troe [66] was preferred in the previous [6] and in the present mechanisms as well as in the models of Burke et al. [33] and of Keromnes et al. [22]. This expression shown in Table 1 is essentially an approximation of the original two-terms rate ($k_\infty = [0.32 \cdot T^{0.56} + 2.9 \cdot 10^4 \cdot T^{-1.7}] \cdot 10^{-11} \text{ cm}^3 \text{ molecule}^{-1}$) for the range 300–2000 K proposed by Troe [66], since the two-term high-pressure rate expression cannot be directly used in the Chemkin interpreter [52]. This rate constant was recommended by Baulch et al. [67] and re-confirmed in the works of Troe and coworkers [68,69]. Fernandes et al. [69] noted that the high-pressure rate derived by Sellevag et al. [64] is about 25% lower than the rate derived by Troe [66] attributing this divergence to the differences in the theoretical model parameters.

Shimizu et al. [43] also preferred the rate constant of Troe [66], yet mistakenly used only the first term of the above expression from Baulch et al. [67]. For the high-pressure rate constant Hong et al. [42] referred on Bates et al. [70] yet used the expression which is more than 6 times higher than the original one. Since the high-pressure rate constant of Troe [66] is about 3 times higher than that of Bates et al. [70] the resulting erroneous rate is about twice as high.

It was emphasized on several occasions [32,33] that the method used in the present and other models to represent different fall-off behavior of particular reaction for different collision partners by writing an individual reaction for each cannot predict appropriate high pressure limit behavior of the mechanism. Each of the expressions reaches the high pressure limit, causing the overall reaction at the high pressure limit to be a sum of true high pressure limit rates. Therefore, Burke et al. [33] proposed 2 combinations of the rate constants to be used for the case when either N₂ or Ar are abundant gases. In this context Hong et al. [42] argued that at typical combustion pressures below 100 atm the fall-off curves are predominantly controlled by the low-pressure rate constant terms. Thus, although the high-pressure limiting terms of the rate constants of reaction (R5) are important to reproduce the fall-off behavior, still the problem of the duplication may arise only at very high pressures for typical combustion temperatures. Therefore in the present model the rate constants for different collisional partners are presented separately.

Low-pressure rate constants for Ar and N₂ as collisional partners [66,70–72] are very close to each other (for each of them) and these data are commonly used in the recent hydrogen mechanisms as outlined in the following and illustrated in Supplemental material (Figs. S1–S5). Most recent experiments of Mertens et al. [73] agree well with the earlier measurements. In the present mechanism the rate expressions recommended by Michael et al. [72] for N₂, Ar and O₂ are adopted. For water, the low-pressure expression obtained by Bates et al. [70] was employed. Termolecular rates at room temperature derived theoretically by Michael et al. [72] are scaled as H₂O:CH₄:O₂:H₂:Ar:He:N₂ = 12:3.5:0.74:1.51:0.49:0.57:1. These collisional efficiencies are preferred in the mechanism because they agree within 20% with the determinations of Ashman and Haynes [74] (H₂O:CO₂:Ar:N₂ = 10.6:2.4:0.56:1) and also with the earlier recommendation [65] (H₂:N₂ = 1.5:1).

Shimizu et al. [43] derived their own expressions for the low pressure rate constants for H₂ and O₂ bath gases, while adopting the rate constants of Michael et al. [72] for argon and other collisional partners. Another mistake in the rate constant for helium is spotted in this mechanism. In the model of Hong et al. [42] the low-pressure expressions were taken from Bates et al. [70] with the exception of O₂ for which the rate of Michael et al. [72] was used. For prevailing N₂ or Ar the corresponding low-pressure rates of Michael et al. [72] were approximated by three-term Arrhenius expressions and employed in the mechanism of Burke et al. [33]. For other components fixed collisional efficiencies relative to the bulk gases are implemented. Keromnes et al. [22] tested recent recommendations of Fernandes et al. [69] who, in turn, used the low-pressure rate constants for Ar and N₂ taken from Michael et al. [72]. Keromnes et al. [22] only adopted the expression for N₂, while for Ar the low pressure rate of Bates et al. [70] was selected, arguing that the rate of Michael et al. [72] leads to very long ignition delays in shock waves at lower temperatures (1000–1200 K). For He the low-pressure rate of Fernandes et al. [69] was multiplied by a factor of 1.5 to fit with negative pressure dependence of the high-pressure flames diluted by He.

Although Troe [66] estimated a reliability of his recommended rates to be about 20%, the discrepancy between different experimental data [70–75] and new expressions proposed and implemented by Shimizu et al. [43], Burke et al. [33] and Keromnes et al. [22] is still about 30–40% as illustrated in the Supplemental material (Figs. S1–S5). In attempt to reconcile different datasets, Turanyi et al. [61] sought for optimal Arrhenius parameters for reactions (R5) and (R8) using direct and indirect measurements. This procedure substantially reduced uncertainty limits for both reactions. The low-pressure rate constant of reaction (R5) proposed for Ar is in very good agreement with the literature, however, the new optimized rate constant for N₂ is significantly higher than other literature expressions at temperatures above 1000 K. This probably is a consequence of simultaneous fitting to direct and indirect measurements performed in Ar or N₂ in assumption of a constant temperature-independent collisional efficiency of Ar relative to N₂.

In the previous version of the hydrogen mechanism [6] reaction



was written in reverse direction. For hydrogen peroxide decomposition, Shimizu et al. [43] adopted high-pressure and low-pressure rates from Baulch et al. [67] and from Kappel et al. [76], respectively. Hong et al. [42] preferred high-pressure rate from theoretical study of Sellevag et al. [77] noting that this rate is much higher than the rate derived in the experiments of Kappel et al. [76]. For the low-pressure rate Hong et al. [42] relied on their own experiments [44,45] performed at relatively low temperatures (1000–1460 K). Troe [78] recently accomplished detailed analysis of the temperature and pressure dependence of H₂O₂ decomposition and his recommendations have been adopted by Burke et al. [33], Keromnes et al. [22] and in the present work. Subsequent experiments [79] performed in He were found to be several dozen percent lower than results of Sellevag et al. [77] and of Troe [78] at high pressures, and by a factor of 2.2–2.8 higher at low-pressure limit. Based on these observations the uncertainty of the rate constant is evaluated as shown in Table 1.

Each new or updated kinetic mechanism for hydrogen includes extended discussion of the chain-branching reaction



as one of the most sensitive and important in combustion models. Several years ago the most popular choices were either the rate constant recommended by Baulch et al. [67] or by Pirraglia et al. [80]. For example, Mueller et al. [81], Ó Conaire et al. [9], and Shimizu et al. [43] preferred the rate expression of Pirraglia et al. [80].

Recent highly accurate and reconciling measurements of Hong et al. [46] are nowadays widely accepted. The uncertainty of this rate constant shown in Table 1 is mostly indicative: it varies depending on the temperature range as discussed by Turanyi et al. [61] and Miki et al. [82].

Baulch et al. [67] estimated the uncertainty of the rate constant of reaction



to be of a factor of 1.5. It was realized long ago, e.g., in [83], that it has strong non-Arrhenius dependence supported by the measurements at high [84] and at low [85] temperatures. Although parametric presentation of the rate constants used in different models is different, they are all within the evaluated uncertainty range. Shimizu et al. [43] used the rate calculated by Michael [86], while Hong et al. [42] preferred results of Wooldridge et al. [83]. The expression proposed by Baulch et al. [67] is almost indistinguishable from the expression of Wooldridge et al. [83] and it is accepted in the present work and by Burke et al. [33]. Keromnes et al. [22] used reverse reaction and the rate of Sutherland et al. [84]. Recent measurements of this rate constant [79] from room temperature up to 834 K confirmed a minimum in the temperature dependence around 500 K and were found to be in good agreement with the shock tube studies [83,84]. Sangwan and Krasnoperov [87] approximated their own data [79] and previous measurements [83,85] by a two-term expression which is systematically lower than recommendation of Baulch et al. [67] by about 10–15%, which is well within the evaluated uncertainties for this reaction. Most recent ab initio calculations of this rate constant [88] are in agreement with majority of experimental data.

Reliable measurements of the rate constant of reaction



are only available at low temperatures. Baulch et al. [67] adopted expression of Atkinson et al. [89] recommended within 220–400 K and extended it to the higher temperatures with uncertainty of 1.4 at 220 K rising to a factor of 3 at 1000 K. Konnov [6] noted that anomalous behavior of the rate constant (positive activation energy at high temperatures and rate constant minimum between 500 and 1500 K) cannot be excluded, yet adopted recommendations of Atkinson et al. [89]. It has also been used in the model of Hong et al. [42]. Shimizu et al. [43] and Keromnes et al. [22] preferred temperature – independent rate suggested by Baulch et al. [65]. Burke et al. [33] used theoretical expression derived by Fernandez-Ramos and Varandas [90] and scaled it down by a factor of 0.6 to match low-temperature measurements [89]. This expression has its minimum around 400 K, and it is accepted here with an estimated uncertainty of a factor of 2 at high temperatures.

From the three channels of the reaction between H atoms and HO₂ radicals Burke et al. [33] and Keromnes et al. [22] included only two



excluding reaction



In the previous model [6] the rate constant of reaction (R12) was taken from Baulch et al. [91] and slightly scaled up to fit room temperature evaluation of the IUPAC [92]. The rate constants of reactions (R13) and (R14) recommended by Baulch et al. [67] were also adopted in that mechanism. These rate constants were based on the recommendations of the IUPAC [92] that made extrapolation to typical combustion temperatures rather uncertain, therefore they were revised in the present model. For reaction (R12), the rate

derived by Mueller et al. [81] was preferred here and in other mechanisms compared in Table 1. It was obtained using room-temperature data of Keyser [93] and Sridharan et al. [94] and recalculated data of Baldwin et al. [95] at 773 K.

In the same model [6] the channel forming H₂O + O was included arguing that it is kinetically different from that forming OH + OH. This reaction (with numbering (–R13) in the present work) was also implemented in the mechanism of Hong et al. [42]. Shimizu et al. [43] recalculated experimental data of Baldwin and Walker [96] at 773 K and using IUPAC data at room temperature proposed their own expression for the rate constant of reaction (–R13) which is close to recommendation of Baulch et al. [67]. Burke et al. [33] performed extensive analysis of the possible branching ratios of reactions (R12)–(R14). Acknowledging previous theoretical studies of Karkach and Osheroov [97] and of Mousavi-pour and Saheb [98] as qualitatively informative, they reinvestigated several possible reaction channels with high level ab initio TST calculations. It was found that formation of H₂O + O (1D) constitutes less than 4%, and formation of H₂O + O (3P) less than 10% of the total H + HO₂ flux at temperatures from 300 to 2500 K. Burke et al. [33] attempted different branching ratios for formation of the ground state O (3P) and concluded that, with the total (*k*₁₂ + *k*_{–13}) fixed, the impact of reaction (–R13) on the model predictions is negligible if the ratio is below 10%.

Available measurements of the branching ratio are limited to room and lower temperatures. Keyser [93] analyzed earlier experiments and from his measurements found that the fraction of H₂O + O formation in reaction between H atoms and HO₂ radicals is 0.02 ± 0.02, which makes the use of these data for extrapolation to higher temperatures very unreliable. Therefore in the present work this channel is written in reverse direction as



with the rate constant calculated by Conforti et al. [99]. Their modeling results were preferred to other works [97,98,100] since the rate constant of the channel



also calculated in this work is very close to the recommendations of Baulch et al. [67]. Moreover, Conforti et al. [99] found good agreement with molecular beam measurements [101] of the relative cross section in reaction (R13). Due to significant disagreement of the theoretical studies [33,97–100] the uncertainty of this rate constant is evaluated to be of a factor of 5.

In the previous model [6] the recommendation of Baulch et al. [67] for the rate constant of reaction (–R14) was adopted. Interestingly, this recommendation was limited to the range 250–1000 K and did not take into account the results of Michael et al. [102], which are implemented in many contemporary models including those of Shimizu et al. [43] and of Hong et al. [42]. Burke et al. [33] and Keromnes et al. [22] scaled down the expression of Michael et al. [102] multiplying it by a factor of 0.75 or 0.7, respectively. Owing to this, the uncertainty of the rate constant is evaluated to be still of a factor of 1.5.

The second channel of initiation in hydrogen + oxygen mixtures

$$\text{H}_2 + \text{O}_2 = \text{OH} + \text{OH} \quad (\text{X1})$$

is of negligible importance as discussed elsewhere, e.g., [6,33], and can be excluded from the model. This and other reactions neglected in the present mechanism are listed in Table 2.

Table 2
Reactions not included in the mechanism.

	Reaction	A	n	E _a	Ref.
X1	H ₂ + O ₂ = OH + OH	2.04E+12	0.44	69,155	[97]
X2	HO ₂ + HO ₂ + M = H ₂ O ₂ + O ₂ + M	6.84E+14	0	–1950	[92]
X3	O + OH + M = HO ₂ + M	1.0E+15	0	0	[123]

All recent publications presenting hydrogen combustion models discuss the role and the choice of the rate constant of reaction



For a long time there was a controversy between the rate constant of Keyser [103] measured within 250–400 K and anomalous behavior (the deep and unusually narrow rate constant minimum close to 1250 K) of this reaction observed by Hippler et al. [104] and subsequently confirmed but at notably shifted temperature by Kappel et al. [76]. Srinivasan et al. [105] found no evidence for temperature dependence of the rate constant from 1237 to 1554 K. Shimizu et al. [43] accepted earlier recommendation of Baulch et al. [65]. The temperature dependence found by Keyser [103] was adopted in the mechanisms of Burke et al. [33] and of Keromnes et al. [22] with 15% reduction in the latter case. Hong et al. [47] measured this rate constant in the range 1600–2200 K concluding that the expression of Keyser [103] can be extended from low up to these high temperatures and implemented it in their model [42]. Most recently Burke et al. [106] re-interpreted experimental measurements for the H_2O_2 decomposition including those indicating deep rate constant minimum [76,104]. Guided by ab initio calculations their analysis showed that the rate constant of reaction (R15) does exhibit weak temperature dependence with a shallow minimum around 1000 K. Simultaneously Hong et al. [48] performed dedicated experiments between 1070 and 1280 K and came to the same conclusion. These two reconciling studies are in remarkable agreement; therefore the expression of Hong et al. [48] is adopted in the present model.

The rate constant obtained by Hippler et al. [107] for reaction

$$\text{HO}_2 + \text{HO}_2 = \text{H}_2\text{O}_2 + \text{O}_2 \quad (\text{R16})$$

has been recommended by Baulch et al. [67,91] and adopted in many combustion mechanisms, e.g., [42,33]. Kappel et al. [76] improved the accuracy of their measurements and extended the temperature range of the new rate expression that was adopted by Konnov [6] and kept in the present work. This rate constant was very recently reconfirmed by Hong et al. [48]. Keromnes et al. [22] in their model adjusted the high-temperature term of the two-term rate expression of Hippler et al. [107] essentially approaching that of Kappel et al. [76].

To investigate a possible role of the pressure dependence of the HO_2 self-reaction, Konnov [6] included reaction



with the rate constant suggested in the review of the IUPAC [92]. This reaction has been recently revisited in theoretical work of Zhou et al. [108]. They found qualitative agreement with the experiments of Kircher and Sander [109] used in the IUPAC evaluation [92], yet the effects of pressure were found less pronounced and negligible at all pressures below 10 atm above 700 K. Therefore this reaction was not included in the present mechanism (see Table 2). One should note that the pressure-dependent part of this reaction is incorrectly interpreted in the model of Shimizu et al. [43] mixing pressure-dependent and independent terms in a Troe-type fall-off rate constant formulation.

Tsang and Hampson [110] and Baulch et al. [67] recommended largely different rate constants for two reactions



due to very limited and controversial experimental data. Several theoretical works [111–113] attempted to resolve this controversy. Ellingson et al. [113] demonstrated that by adjusting reaction barrier height the calculated rate coefficients can be fitted to different experimental results. Higher rate constants derived by Ellingson

et al. [113] (so-called set II) were preferred by Shimizu et al. [43] and adopted in the present work. Keromnes et al. [22] also accepted calculated rate constant of reaction (R17), keeping expression of Tsang and Hampson [110] for reaction (R18). It should be noted that the same calculated values [113] were approximated by different Arrhenius expressions in these works [22,43]. For reaction (R17) Hong et al. [42] used expression from the GRI-mech [15], which was essentially a fit of the data reported in Tsang and Hampson [110] to a T^2 temperature dependence, and the rate constant from Baulch et al. [67] for reaction (R18). Burke et al. [33] consistently used recommendations of Tsang and Hampson [110].

The rate constant of reaction



measured and analyzed by Hippler et al. [104] and reconfirmed by Kappel et al. [76] has been used in many models including those of Konnov [6] and Shimizu et al. [43] before recent study of Hong et al. [45]. Improved diagnostics allowed for better accuracy of the rate constant measurements, which are adopted in the recent mechanisms [22,33,42] and in the present work.

3. Mechanism validation

No attempt to adjust the reaction rate coefficients was made in the present work. In the following figures, the present model is referred to as “current mechanism” and plotted as solid lines. The performance of the previous version of the Konnov mechanism [6] and the best contemporary model of Keromnes et al. [22], referred to as “Konnov 2008” and “Keromnes 2013” and plotted in dashed and dash-dotted lines, respectively, is shown for comparison. A selected set of validation cases is presented, with a particular attention on those which illustrate the improved performance of the present mechanism. A comprehensive set of validation runs can be found in the [Supplemental material](#). Modeling range covers ignition experiments in shock tubes and rapid compression machines (RCM) from 900 to 2700 K and from sub-atmospheric pressures up to 87 atm; hydrogen oxidation in a flow reactor at temperatures around 900 K from 0.3 up to 15.7 atm; flame burning velocities in hydrogen + oxygen + inert mixtures from 0.35 up to 25 atm at room and elevated temperatures up to 443 K; hydrogen flame structure at pressures from 0.05 to 10 atm.

3.1. Modeling details

The modeling was performed using Chemkin Software Release 10112 [124]. For the simulations of the jet-stirred reactor studies, the PSR Reactor Model was used. The Closed Homogeneous Reactor Model was used for the simulations of flow reactor, RCM and shock tube data. For the case of the flow reactor simulations, constant pressure and enthalpy were assumed and the modeled species profiles were shifted to the position where the mole fraction of H_2 equals 50% of the initial value from the corresponding experimental study. For the simulations of the species profiles recorded in shock tubes and ignition delay times, constant volume and internal energy were assumed. In the modeling, the original criteria for ignition from the corresponding studies were reproduced, unless otherwise specified, except for the experiments where OH^* emission was recorded. In these cases, for either maximum concentration of OH^* or its first derivative, the ground state OH gradient was used as the ignition criterion, since the current mechanism does not include excited radicals. Even though the model of Keromnes et al. [22] includes OH^* , for the purpose of fair comparison to the current mechanism, the same ignition criteria were used for all three models considered. Ignition delay times in RCM were calculated applying the experimental volume history. For the simulation

of the burning velocity and flame structure studies, PREMIX code of Chemkin was used. The modeling was performed with multicomponent transport coefficients and thermal diffusion option. Adaptive mesh parameters were GRAD 0.005 and CURV 0.1 for the free propagating model, resulting in typical number of grid points around 1500. The simulations of the flame structure were performed with experimental temperature profiles approximated with a spline function. Adaptive mesh parameters were GRAD 0.005 and CURV 0.5, resulting in typical number of grid points above 1000.

3.2. H_2 oxidation in JSR and flow reactors

Figure 2 presents simulations of the jet-stirred reactor data of Dayma and Dagaut [125] for $H_2 + O_2 + N_2$ mixtures at 10 atm. This type of validation cases was not covered previously [6]. The exact initial reactant concentrations were taken from the corresponding figure of [125] at the temperature where no reaction occurs, the same was done in [125], since the initial H_2O concentration in Fig. 2 is non-negligible. Residence time for the measurements was 1 s.

Figure 3 shows the flow reactor data of Mueller et al. [81] for $H_2 + O_2 + N_2$ mixture at 3.02 atm. The experimental points are plotted with uncertainties specified in [81]. The modeling data were shifted to the position where the mole fraction of H_2 equals 50% of the initial value. It should be noted that the results from [81] were included in the validation set for the previous version of the mechanism [6], and the discrepancy was found to be significant. For the current version, the agreement was found to be very good. This is illustrated by Figs. S11–S14 in the Supplemental material, where additional validation runs for the JSR and flow reactors can be also found (Figs. S6–S15).

3.3. Oxidation and decomposition in shock waves

The previous version of the mechanism [6] was validated against a wide range of ignition delay times measured in shock tubes, however, other types of experiments were not covered. In the current work, the mechanism performance was tested against several studies where species profiles were recorded in shock tubes [45–47,126]. As an example, a profile of H_2O from H_2O_2 decomposition study of Hong et al. [45] is shown in Fig. 4, while other results can be found in the Supplemental material (Figs. S16–S21). For the conditions of [45,47], there was a significant improvement compared to the previous model [6]. As dis-

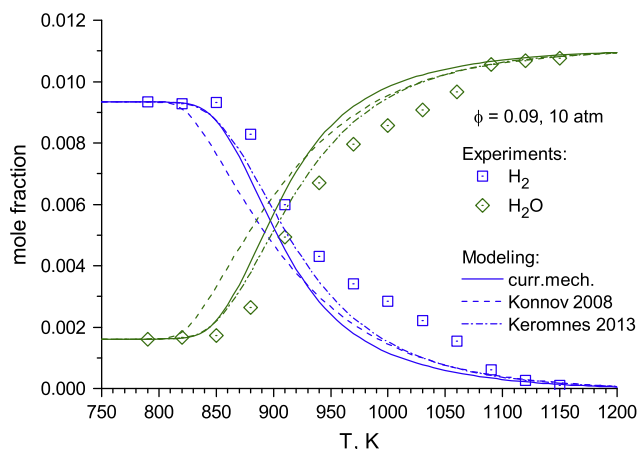


Fig. 2. Profiles of H_2 and H_2O for H_2 oxidation in JSR [125]; $H_2 \approx 1\%$; $\phi \approx 0.1$; $p = 10$ atm.

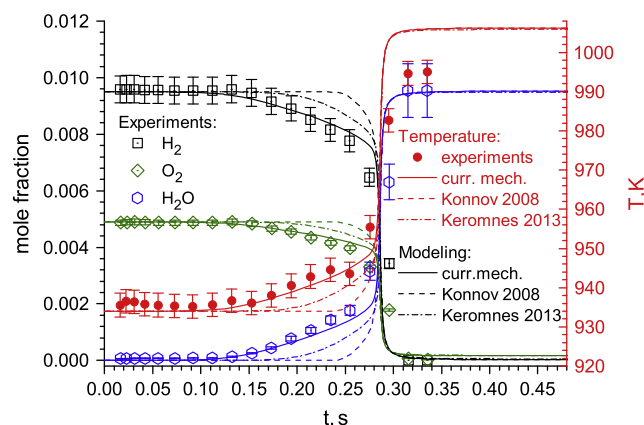


Fig. 3. Profiles of major species and temperature for a stoichiometric mixture ($H_2 = 0.95\%$, $O_2 = 0.49\%$) at 3.02 atm [81]. The mixture is diluted by N_2 .

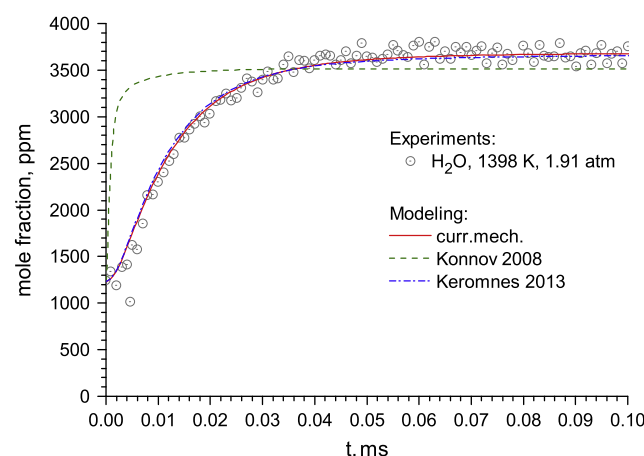


Fig. 4. Profile of H_2O in shock wave decomposition of H_2O_2 [45]. The initial conditions are: $H_2O_2 = 2540$ ppm, $H_2O = 1234$ ppm, $O_2 = 617$ ppm, $T = 1398$ K, $p = 1.91$ atm. The mixture is diluted by Ar.

cussed in Section 2.2, the study of Hong et al. [47] was aimed at determination of the rate constant of reaction (R15), and in the current mechanism, the expression from the later study of Hong et al. [48] is adopted. The experimental data of Hong et al. [45] allowed for determination of the rate constants of reactions (R20) and (R6). Hong et al. [42] compared the performance of several kinetic mechanisms for the conditions of [45] and demonstrated that the previous version of the mechanism [6] had a major discrepancy in terms of species profiles prediction (see Fig. 4 and Figs. S19–S21). Indeed, the rate constant of reaction (R20) used in [6], which was taken from [104], lied far from the expression determined in [45]. In the current model, the expression from [45] was directly implemented for reaction (R20). For the reaction (R6), a rate constant of Troe [78] was applied in the current mechanism, however, it was noted in [78] that the expression is close to the measurements of Hong et al. [45]. Hence, an excellent agreement between the data of [45] and the performance of the current model was observed in the present study.

3.4. Ignition delays

Figure 5 presents the modeling of ignition in a rapid compression machine (RCM) using the experimental data of Mittal et al. [127]. The volume history for the measurements was provided online [128]. The ignition delay times were defined as the interval

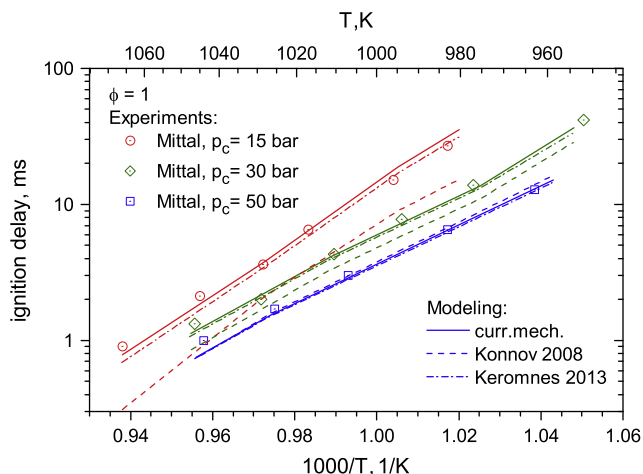


Fig. 5. Ignition delay times for stoichiometric H_2 mixtures at compression pressures $p_c = 15\text{--}50$ bar [127]. The initial reactant concentrations are $\text{H}_2 = 12.5\%$, $\text{O}_2 = 6.25\%$, $\text{N}_2 = 18.125\%$, $\text{Ar} = 63.125\%$.

from the end of compression cycle to the point with maximum pressure gradient and they are plotted as a function of the compression temperature.

Figure 6 contains the measurements conducted in a shock tube by Keromnes et al. [22], which constituted a part of the validation set for the mechanism [22]. The experimental ignition delay times are based on $[\text{OH}^*]_{\text{max}}$ and simulated with $(d[\text{OH}]/dt)_{\text{max}}$ in the present work.

The measurement range of data from [22] includes temperatures around and below 1000 K. Hence, the corresponding ignition delay times are of the order of several ms. For such long delay times, the self-ignition processes are generally affected by the facility-related pressure and temperature rise, which accelerate the ignition. The authors of [22] took the gas-dynamic effects into account by recording pressure and temperature histories of non-igniting mixtures and coupled them to the kinetic effects by using in-house written extension to Chemkin II. The gas-dynamic effects can result in a large discrepancy between the modeling and experiments, as evident from Fig. 6, opposed to the simulation results from [22], where the agreement was found to be very good.

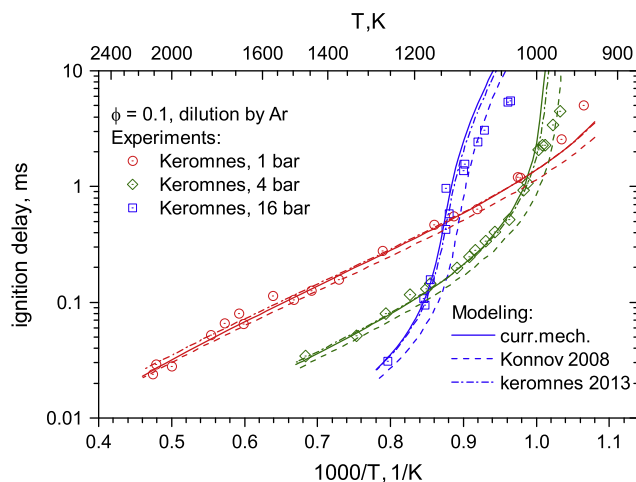


Fig. 6. Ignition delay times for lean ($\phi = 0.1$) $\text{H}_2 + \text{O}_2 + \text{Ar}$ mixtures at different pressures [22]: $p_0 = 1$ bar (red circles), $p_0 = 4$ bar (green diamonds), $p_0 = 16$ bar (blue squares). The initial concentrations are $\text{H}_2 = 0.81\%$, $\text{O}_2 = 4.03\%$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

However, for the shorter ignition delays of Fig. 6, it can be seen that the current mechanism has an improved performance as compared to the previous model [6]. Additionally, if pressure history is provided together with the published ignition delay data, as, for example, in the study of Pang et al. [129], the current mechanism is able to adequately reproduce long ignition delays as shown in Fig. S37 in the Supplemental material. Also, more validation cases for shock-wave and RCM ignition are presented in Figs. S22–S41.

3.5. Laminar burning velocity and flame structure of H_2 mixtures

Figures 7 and 8 show the laminar burning velocity of $\text{H}_2 + \text{air}$ mixtures at 1 atm [7,8,130–146] at room and elevated temperatures, respectively. It can be seen that both the current model and the previous version [6] agree well with the most recent experimental data in a wide range of equivalence ratios except very lean and very rich flames. As will be shown in Section 4, in the range where a good agreement with the burning velocities was observed, the temperature exponent α (see Eq. (1)) also agrees well with the experimental values used for its extraction, e.g., from the recent study of Krejci et al. [145]. In very lean mixtures (ϕ

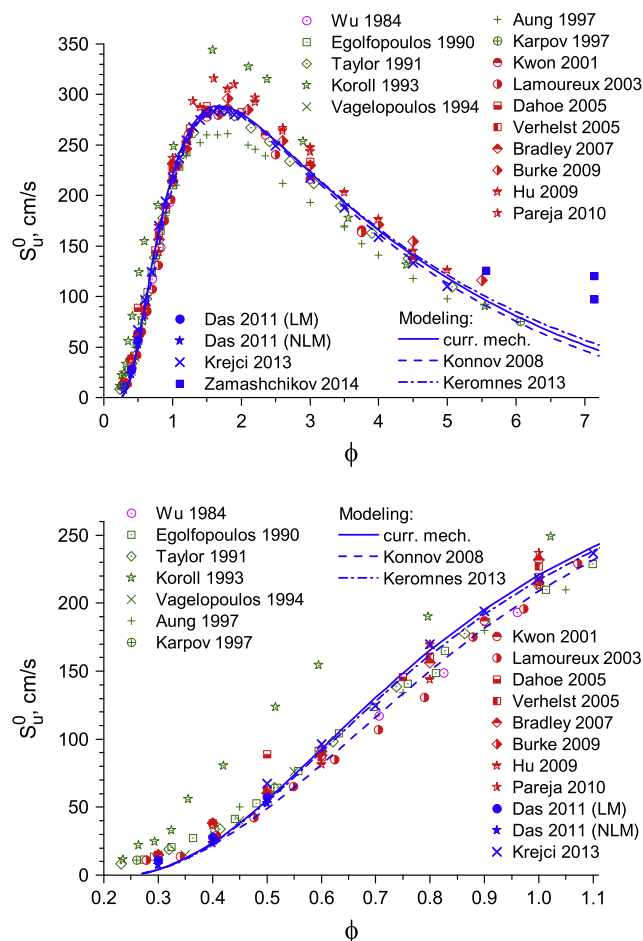


Fig. 7. Laminar burning velocity of $\text{H}_2 + \text{air}$ flames at standard conditions ($T = 298$ K, $p = 1$ atm) for $\phi = 0.25\text{--}7$ (top) and $\phi < 1$ (bottom). The source of experimental data: before 1990 (purple symbols) – Wu and Law [130]; from 1990 to 2000 (green symbols) – Egolfopoulos and Law [131], Taylor [132], Koroll et al. [133], Vagelopoulou et al. [134], Aung et al. [135], Karpov et al. [136]; from 2000 to 2010 (red symbols) – Kwon and Faeth [137], Lamoureux et al. [138], Dahoe [139], Verhelst et al. [7], Bradley et al. [140], Burke et al. [141], Hu et al. [8], Pareja et al. [142]; after 2010: Das et al. (LM – linear model, NLM – non-linear extrapolation) [143,144], Krejci et al. [145], Zamashchikov et al. [146]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

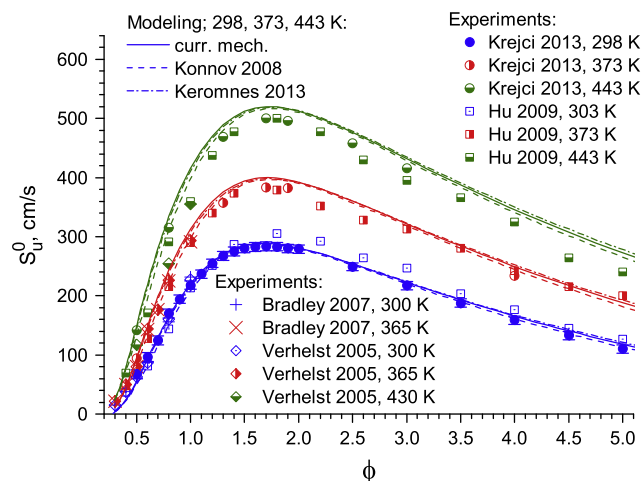


Fig. 8. Laminar burning velocity of H_2 + air flames at standard conditions (blue color) and elevated temperatures (≈ 373 K – red; ≈ 443 K – green) for $\phi = 0.3$ – 5 . The source of experimental data: Krejci et al. [145], Hu et al. [8], Bradley et al. [140], Verhelst et al. [7]. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

< 0.5) the experimentally measured burning velocities possess higher relative scattering and disagree with the model predictions both in terms of burning velocity, and consequently, in values of α .

In the **Supplemental material**, additional validation cases at atmospheric, elevated and low pressures are presented for H_2 + air mixtures and for the mixtures diluted by N_2 , Ar, He and H_2O (Figs. S42–S57). Helium was introduced into the current version of the model, and its predictive ability for He-diluted flames can be assessed from Figs. S49, S52 and S54–S57.

Figure 9 shows major and minor species of low-pressure $\text{H}_2 + \text{O}_2 + \text{Ar}$ flame from the experiments of Vandooren and Bian [147]. The simulations were conducted with the experimental temperature profile, which is also shown for reference. No shift of species profiles relative to temperature profiles was performed in either experimental or modeling data. It can be seen that the performance of the three models is identical and they agree well with experiments in terms of major species profiles. There is a considerable disagreement in minor species prediction, however, the accuracy of the molecular beam sampling technique for minor species determination and of the equilibrium calibration, used in [147], can be questioned due to important flame distortion by a probe [148]. An extended set of validation runs of the flame structure at atmospheric and elevated pressures can be found in the **Supplemental material** (Figs. S58–S61).

3.6. Discussion

Previous hydrogen combustion model [6] was evaluated [40] to be in the short list of the best mechanisms which also includes the models of Saxena and Williams [41], Hong et al. [42], 2 models originating from Princeton [32,33] and from NUIG [9,22]. In some cases, however, predictions of the Konnov model [6] significantly disagreed with available, most recent measurements in shock tubes (see Fig. 4 and Figs. S19–S21) or in flow reactors (Fig. 3 and Figs. S11–S14). These deficiencies were addressed in the present work and resolved often thanks to implementation of the rate constants measured by Hong et al. [44–48].

Direct comparison of the rate constants adopted in the Konnov mechanism [6], other models published since 2008 [22,33,42,43] and accepted in the present work (see Table 1) clearly demonstrates that consensus on at least some of the hydrogen reactions can be reached without rate constant adjustment. Validation

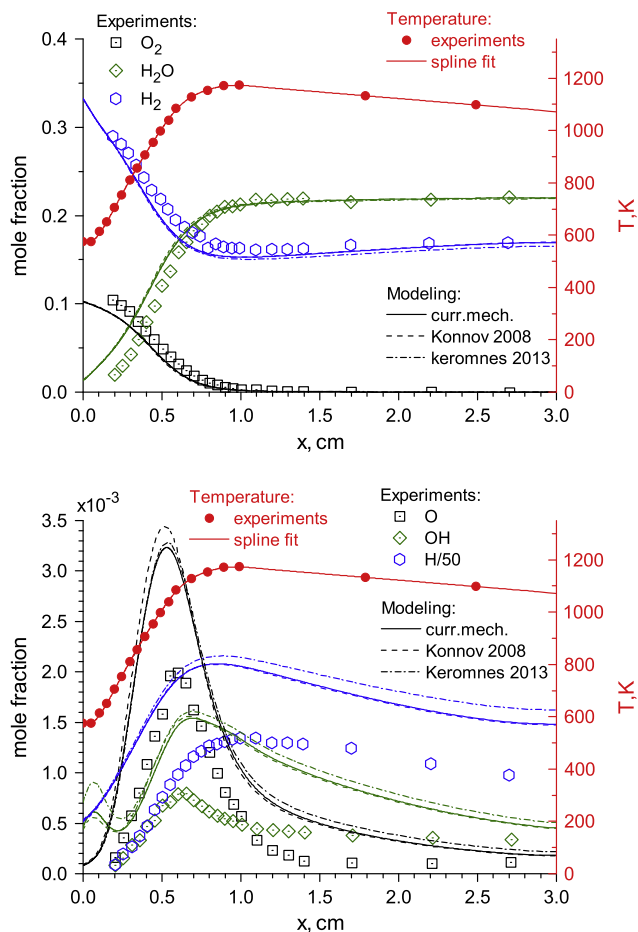


Fig. 9. Profiles of major (top) and minor (bottom) species in rich ($\phi = 1.93$) low pressure (4.75 kPa) $\text{H}_2 + \text{O}_2 + \text{Ar}$ flame [147] ($\text{H}_2 = 39.7\%$, $\text{O}_2 = 10.3\%$, $\text{Ar} = 50\%$). Initial temperature is 576 K and the corresponding unburned gas velocity $u_0 = 253$ cm/s.

examples shown above and in the **Supplemental material** illustrate that performance of the new hydrogen mechanism is close to that of the mechanism of K  romn  s et al. [22], which was shown to be the best among other models [40]. For the fair comparison, analysis of the effects of temperature on the adiabatic burning velocities of hydrogen flames was performed using both kinetic schemes as described in the following.

4. Temperature dependence

Temperature dependence of the laminar burning velocity in the form of Eq. (1) is important for several reasons: first, it can be used in CFD codes to correlate the burning velocity at the elevated temperatures not accessible for laboratory measurements. A number of examples of such correlation studies are shown and discussed in the Introduction. Moreover, the temperature exponent α can be used as independent parameter for detailed kinetic model validation, since the burning velocity and its temperature dependence can be governed by different sets of reactions. Finally, adopting Eq. (1) for correlation of the burning velocity data allows to describe sometimes extensive data by only two parameters, $S_{u,0}$ and α , and the quality of correlation can help to indicate possible systematic errors in the measurements.

The new mechanism was employed to investigate the temperature dependence of the laminar burning velocity of H_2 flames. The conditions of Fig. 1, i.e. the standard conditions of $p = 1$ atm and $T_0 = 298$ K, where the majority of available literature data were

obtained, will be first discussed. Apart from the studies shown in Fig. 1, where the power exponent α was derived directly, a number of works [8,133,140,143–145,149] presented burning velocities of H_2 + air flames at room and elevated temperatures without evaluation of the power exponent in the form of Eq. (1). For these studies, the power exponent α was extracted with a least-square fit procedure. All available and extracted values of α from the experimental data are shown in Fig. 10.

The extracted power exponents in Fig. 10 are plotted with uncertainties, which were evaluated based on the accuracy of the burning velocity measurements. As calculated with the least square model, α is a function of individual burning velocities at each temperature, $S_L^{T_i}$, i.e. $\alpha = f(S_L^{T_1}, \dots, S_L^{T_n})$, therefore, its uncertainty can be estimated with the error propagation rule. For a simple linear regression in logarithmic coordinates, the uncertainty in α can be derived as:

$$\Delta\alpha = \frac{\left(\sum_i \left[\left(\ln \frac{T_i}{T_0} - \overline{\ln \frac{T_i}{T_0}} \right) \cdot \frac{\Delta S_L^{T_i}}{S_L^{T_i}} \right]^2 \right)^{0.5}}{\sum_i \ln^2 \frac{T_i}{T_0} - n \cdot \left(\overline{\ln \frac{T_i}{T_0}} \right)^2} \quad (3)$$

where n is the number of data points, $\Delta S_L^{T_i}$ is the uncertainty of the burning velocity at temperature T_i and $\overline{\ln \frac{T_i}{T_0}}$ is mean logarithmic normalized temperature:

$$\overline{\ln \frac{T_i}{T_0}} = \frac{\sum_i \ln \frac{T_i}{T_0}}{n} \quad (4)$$

The uncertainties $\Delta S_L^{T_i}$ were taken from the corresponding studies ([133,140,143–145] and from [150] for Verhelst et al. [7]) or evaluated at 95% confidence level from the burning velocity scat-

tering if the uncertainty was not presented directly (for H_2 + O_2 mixtures discussed later in the text). For the experiments of Koroll et al. [133] and for the point $\phi = 0.45$ of Das et al. [143,144], the burning velocities had to be interpolated from different equivalence ratios, since they did not match the elevated temperature values, and this additional uncertainty was added to $\Delta S_L^{T_i}$.

For the study of Hu et al. [8], there was no information available regarding $\Delta S_L^{T_i}$, thus $\Delta\alpha$ was estimated as a standard deviation of the least-square model coefficient from the total sum of the residuals. The same was done for the point $\phi = 0.8$ of [7], where the fitted with Eq. (1) burning velocities stood outside the experimental uncertainty interval $S_L^{T_i} \pm \Delta S_L^{T_i}$. For all other cases and studies, the burning velocities interpreted with Eq. (1) were within $S_L^{T_i} \pm \Delta S_L^{T_i}$.

For the modeling with the current mechanism, the power exponent α was obtained from a least-square fit to the simulated data in the range of temperatures 250–500 K. As will be shown below, the value of α does depend on the fitted temperature range, which also means that the power law in the form of Eq. (1) with constant α is not valid for a wide temperature intervals. However, for 250–500 K such correlation is still reasonable, and it corresponds to the range of the majority of the H_2 + air burning velocity measurements. Also shown for comparison in Fig. 10 is the performance of the most recent H_2 model of Keromnes et al. [22] at the same conditions.

The two models, as well as the previous version of the mechanism [6], were found to give almost identical results. Comparing the calculations to the experimental data, it can be observed that they are in a reasonable agreement with the most recent measurements of Krejci et al. [145]. In the original article, the authors have supported their measurements with the simulations of the burning velocity with the model of Keromnes et al. [22], and the agreement was found to be very good. The data of [145] also possess less scattering for the $\alpha(\phi)$ dependence than the measurements of Hu et al. [8].

However, in the very lean mixtures ($\phi < 0.5$), not covered in [145], a solid conclusion cannot be drawn due to the large discrepancies in the experimental data. Notably, at $\phi = 0.3$ there are three measurements available, from Verhelst et al. [7], Bradley et al. [140] and Das et al. [143,144], which differ by about a factor of 5. Also, all modeling studies shown in Fig. 1 predict a trend of rapid rise of α in lean mixtures, which is supported by the data of Krejci et al. [145] and Das et al. [143,144]. The latter is the only study, which indicates a very high α in a very lean mixture. However, it should be noted that in this case α is based on just two temperature points in a narrow interval of 25 K (298 K and 323 K), though the rise of the burning velocity was considerable. Nevertheless, the results extracted from [143,144] are valuable as a possible evidence for a trend found in the modeling, yet should not be seen as an attempt to determine exact value of α at $\phi = 0.3$.

It should be also noted that Das et al. [143] employed the counterflow technique. In the other studies of very lean H_2 + air flames, Bradley et al. [140] and Verhelst et al. [7], a spherical bomb method was used. For these last two cases [7,140], the flame configuration was unstable due to a high Lewis number at $\phi = 0.3$. This could be seen as one of the possible reasons for the discrepancy.

Comparing the S_L correlation studies [10,12,14,16,21], the equation of Ravi and Petersen [21], which was derived for a wide range of ϕ , was found to be the closest to the present modeling.

As can be seen from Fig. 10, for the H_2 + air mixtures there is a large variety of experimental data, resulting in a large scattering of the extracted values of α . For the H_2 + O_2 + N_2 mixtures with variable composition, the amount of available experiments is limited. Two studies exist, where the laminar burning velocity of pure H_2 + O_2 mixtures was measured from cryogenic temperatures to room conditions: Edse and Lawrence [30] and Zitoun and Deshaies [29]. In the latter, the power exponent α was presented directly. The comparison between the measurements and modeling for

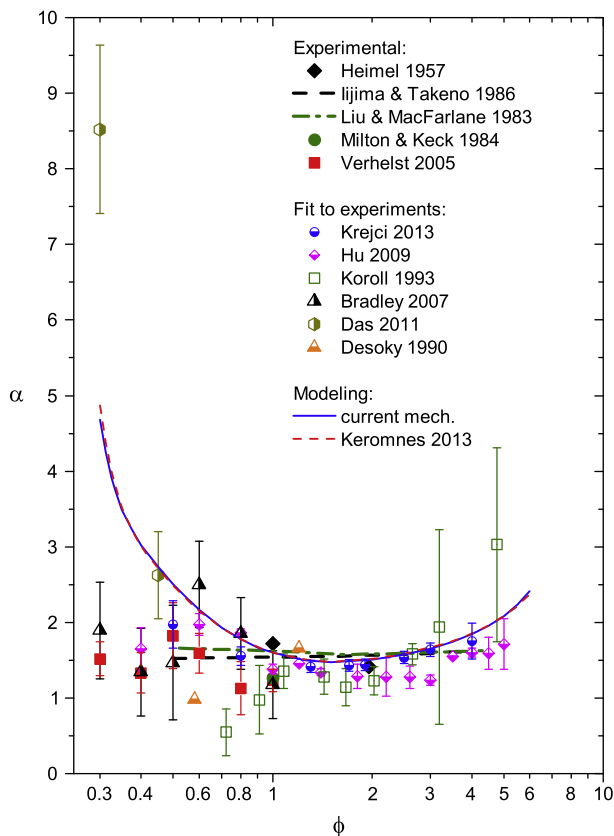


Fig. 10. Power exponent α for H_2 + air flames at standard conditions. Solid symbols and thick lines: experiments [1–4,7], open symbols: fit to experimental data made in the present study [8,133,140,143–145,149], thin lines: modeling.

$\text{H}_2 + \text{O}_2$ mixtures is shown in Fig. 11. The error bars for the values of α for both studies [29,30] were evaluated in the present work from the experimental scattering of the burning velocities.

The investigated temperature range for [30] is about 150 K wider than in [29] and does not match the modeling conditions of Fig. 10 (250–500 K). Therefore, the modeling results in Fig. 11 are shown for three ranges of temperature for fair comparison. It was found that the effect of temperature on the extracted value of α is non-negligible. The rise of α in the rich mixtures is well predicted by the modeling, though the calculated values are somewhat higher than the experimental ones.

As follows from Fig. 11, the power exponent α depends on the temperature range covered. It was observed by Hu et al. [8], who studied the variation of α by extending the temperature range from 300–350 K to 300–950 K. This resulted in the values averaged over a very wide temperature range. In the present study, the variation of α as a function of temperature is presented as $\alpha(T_0)$, where at each central T_0 the fitted interval was chosen to be 100 K: from $T_0 - 50$ K to $T_0 + 50$ K. As an example, the results obtained with the current model for $\text{H}_2 + \text{air}$ (conditions of Fig. 10) and $\text{H}_2 + \text{O}_2$ (Fig. 11) mixtures are shown in Fig. 12.

For $\text{H}_2 + \text{air}$ flames, the variation of α is significant in lean mixtures, while for the mixtures close to stoichiometric, the change of α over the whole 500 K range does not exceed 0.5. Nevertheless, this questions the validity of Eq. (1) as a correlation equation for wide temperature ranges. The variation of α with temperature results in considerable discrepancies between the burning velocities calculated using detailed kinetic mechanisms and the values obtained with Eq. (1), if the latter are extrapolated outside the range where the power exponent was fitted. To illustrate this, in Fig. 13 the laminar burning velocity as a function of temperature of two $\text{H}_2 + \text{air}$ flames, rich $\phi = 1.5$ and lean $\phi = 0.5$, are plotted. The solid lines in Fig. 13 are calculated with the current mechanism, the dash-dot lines are correlations based on the temperature range of 250–500 K. Also shown with the correlation curves are error bars, which illustrate the possible uncertainty in extrapolated burning velocity, if experimental data are used for such kind of extrapolation. The uncertainty in α was estimated from the data of Krejci et al. [145] and is about $\Delta\alpha = 0.1$ for $\phi = 1.5$ and $\Delta\alpha = 0.3$ for $\phi = 0.5$. It can be seen that with increasing temperature, the burning velocity rises faster than the power law of Eq. (1), which can result in significant offset from the real value, as well as in high uncertainty of the value obtained from correlation, if it is extrapolated from the experimental data to higher temperatures.

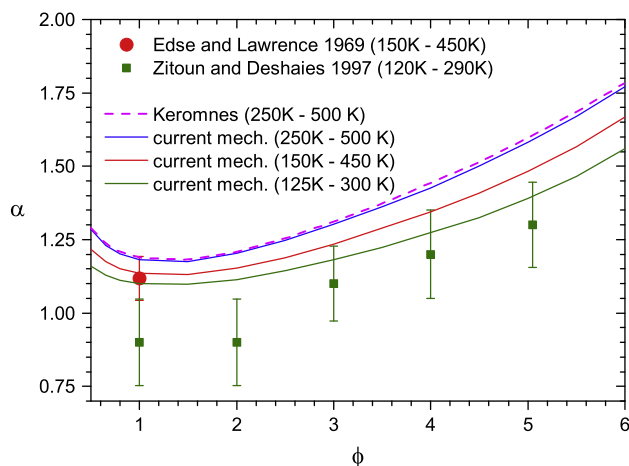


Fig. 11. Power exponent α for $\text{H}_2 + \text{O}_2$ flames at atmospheric pressure. Symbols: experiments [29,30], lines: modeling.

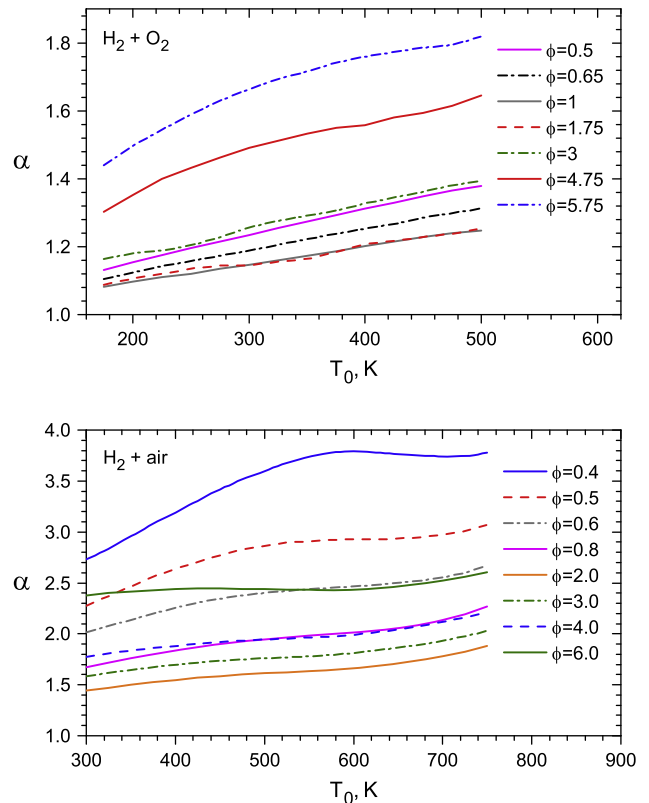


Fig. 12. Effect of equivalence ratio and central temperature on power exponent α for $\text{H}_2 + \text{O}_2$ (top) and $\text{H}_2 + \text{air}$ mixtures (bottom).

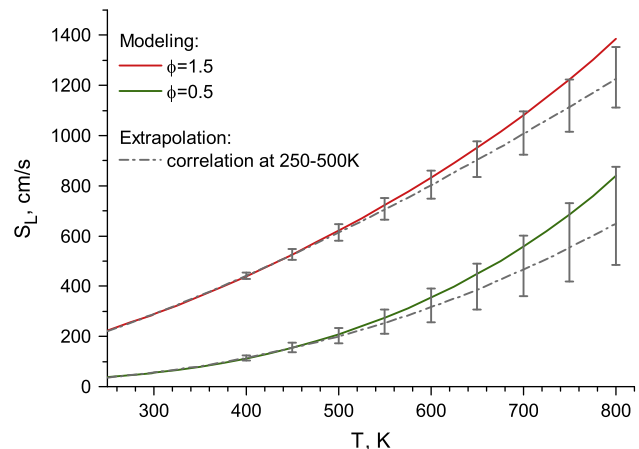


Fig. 13. Laminar burning velocity of $\text{H}_2 + \text{air}$ flames calculated with the current mechanism at $\phi = 1.5$ (red solid line) and $\phi = 0.5$ (green solid line) and obtained with Eq. (1) based on 250–500 K temperature range (gray dash-dot lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The power exponents α for $\text{H}_2 + \text{air}$ mixtures highly diluted with N_2 were evaluated in the study of Paidi et al. [31]. The conditions of this work were modeled with the current mechanism and the model of Keromnes et al. [22] and the results are shown in Fig. 14. The authors used the dilution ratio expressed as $\beta = \text{N}_2^{\text{added}} / (\text{N}_2^{\text{added}} + \text{H}_2)$, which, if converted to the total N_2 fraction in the mixture (both N_2^{air} and $\text{N}_2^{\text{added}}$) results in the dilution of about 73–74% N_2 for the first series ($\phi = 0.8\text{--}1.3$) and about 79.7% N_2 for the second one ($\phi = 0.8\text{--}1.3$). The experimental temperature range of [31] was 300–600 K, so that the modeling results for the conditions of

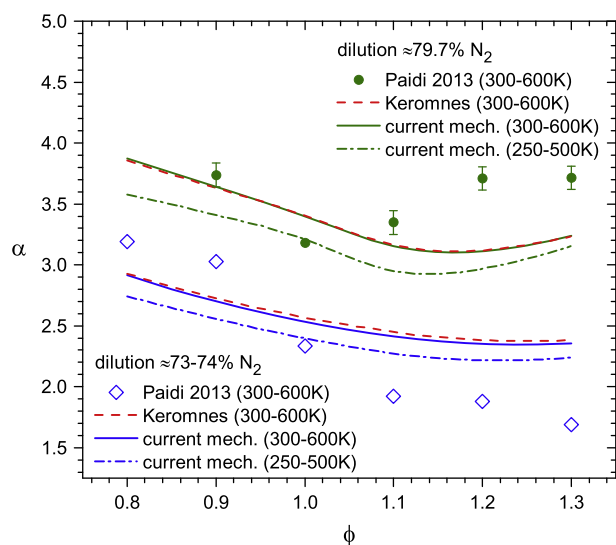


Fig. 14. Power exponent α for $\text{H}_2 + \text{O}_2 + \text{N}_2$ flames at atmospheric pressure and two different dilution ratios. Symbols: experiments [31], lines: modeling.

Fig. 10 (250–500 K) are given for comparison. The error bars were evaluated in the present study based on the stated experimental uncertainty for the burning velocity measurements. The current mechanism and the model of Keromnes et al. match each other, and both models agree well with the experimental points around stoichiometric conditions, with more notable differences in rich mixtures. It should be mentioned that the modeling results are also very close to the results presented in [31] using the Li et al. mechanism [32].

Finally, available results for stoichiometric $\text{H}_2 + \text{O}_2 + \text{N}_2$ mixtures with different amount of N_2 can be combined in one plot. The modeling was done in the range of 250–500 K and the comparison with available literature data is shown in Fig. 15. Apart from the already discussed sources [29–31] and the most recent data from Hu et al. [8] for stoichiometric $\text{H}_2 + \text{air}$ mixture, the results of Kusharin et al. [23] were used. The authors evaluated α at each dilution ratio in a single spherical flame propagation experiment, similar to the work of Milton and Keck [3]. It can be seen that agreement with the models is good, especially for higher dilutions. As was shown before, the rapid rise of α was also observed in the lean $\text{H}_2 + \text{air}$ flames, however, for the case of diluted flames it is clearly supported by experimental observations.

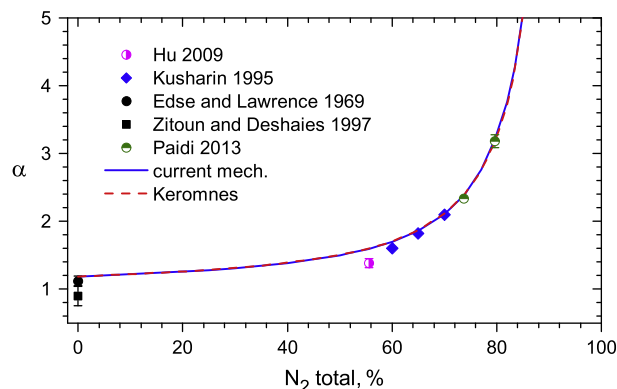


Fig. 15. Power exponent α for stoichiometric $\text{H}_2 + \text{O}_2 + \text{N}_2$ flames at standard conditions and variable N_2 content. Symbols: experiments [8,23,29–31], lines: modeling.

5. Conclusions

A new version of the detailed reaction mechanism for hydrogen was presented. The contemporary choice of the reaction rate constants was provided with the emphasis on their uncertainties. The analysis of the performance of the updated mechanism is presented and compared to the previous version [6] for a wide range of validation cases: jet stirred and flow reactors; oxidation, decomposition and ignition in shock waves; ignition in RCM; laminar burning velocity and flame structure. The deficiencies of the previous version of the mechanism, particularly for the flow reactor and shock tube data, were addressed in the present work and resolved often thanks to implementation of the rate constants measured by Hong et al. [44–48]. Validation examples shown in the present paper illustrate that performance of the new hydrogen mechanism is close to that of the mechanism of K  romn  s et al. [22], which was shown to be the best among other models [40]. No adjustments of the experimental rate constants were required.

Temperature dependence of the laminar burning velocity of hydrogen and the effects of equivalence ratio and dilution on the power exponent α were studied. The contemporary kinetic models for hydrogen combustion are capable to reproduce the experimental data in a wide range of conditions. New experimental measurements in the very lean $\text{H}_2 + \text{air}$ mixtures are required to confirm the quality of the model prediction for these conditions.

A rapid increase of α in very lean and highly diluted mixtures was observed. However, while for the case of diluted $\text{H}_2 + \text{O}_2 + \text{N}_2$ flames an excellent agreement was found between the simulations and experiments, for the lean $\text{H}_2 + \text{air}$ flames, the behavior differed significantly.

The dependence of the power exponent α on the fitted temperature range was studied. The effect is considerable for the power exponent itself, and in terms of burning velocity correlation, it can also result in significant discrepancies, if burning velocity correlation is based on experimental data and used outside the range where the power exponent was fitted. This suggests reconsideration of the power law dependence for the H_2 burning velocity correlation at high temperatures.

Thus, it was shown that the temperature exponent cannot be treated as a constant for hydrogen flames, even to a first approximation. For a given fuel–oxidizer mixture, it strongly depends on equivalence ratio, diluent content and correlation temperature interval. In undiluted H_2 mixtures around stoichiometry α is almost constant and close to 1.5, however, on the sides of the domain, e.g. in lean, rich and highly diluted mixtures, α tends to rise. The same holds for its dependence on initial temperature: α tends to rise with shifting the correlation interval to higher temperatures, especially in lean mixtures.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.combustflame.2014.12.009>.

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